MILITARY SPECIFICATION

HOSE, SYNTHETIC RUBBER, WIRE REINFORCED, AND END FITTINGS, REUSABLE, FOR FLEXIBLE HOSE ASSEMBLIES (FOUR PLIES, SPIRAL-WRAPPED, BRAIDED HIGH PRESSURE SIZES -16 THROUGH -32)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers synthetic rubber hoses reinforced by four plies of spiral wrapped wire in sizes -16 through -32 for high pressure applications.

1.2 Classification. End fittings for hoses covered by this specification shall be sizes -16 through -32 as applicable and shall be of the following types:

Type AS - SAE, 37-degree tube swivel to hose, straight, tube size 1 inch (see figure 1).
Type BS - Gasket seal union, male half to hose, straight, tube size 1 inch (see figure 2).
Type BLS - Gasket seal union, male half to hose, 90-degree elbow, tube size 1 inch (see figure 3).
Type CS - O-ring seal union, male half to hose, straight, tube size 1 inch (see figure 4).
Type CLS - O-ring seal union, male half to hose, 90-degree elbow, tube size 1 inch (see figure 5).
Type DS - 90-degree elbow, hose to hose, hose dash size -16 (see figure 6).
Type CLS - SAE 37-degree tube swivel to hose, straight, tube size 1-1/4 to 2 inches (see figure 7).
Type BO - Gasket seal union, male half to hose, straight, tube sizes 1 inch to 2 inches (see figure 8).
Type BLO - Gasket seal union, male half to hose, 90-degree elbow, tube sizes 1-1/4 to 2 inches (see figure 9).
Type CO - O-ring seal union, male half to hose, straight, tube sizes 1 inch to 2 inches (see figure 10).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4720
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Type CLO - O-ring seal union, male half to hose 90-degree elbow, tube sizes 1-1/4 to 2 inches (see figure 11).
Type DO - 90-degree elbow, hose to hose, hose dash sizes -20 through -32 (see figure 12).
Type UO - 180-degree elbow, hose to hose, hose dash size -24 (see figure 13).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500).
QQ-N-288 - Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.

MILITARY

MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-T-16420 - Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706).
MIL-L-17731 - Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-H-24135 - Hose, Synthetic Rubber, Wire Reinforced, and End Fittings, Reusable, for Flexible Hose Assemblies, General Specification for.

(Copies of specifications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AMERICAN IRON AND STEEL INSTITUTE (AISI)
Steel Products Manual - Stainless and Heat Resisting Steels.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)
AMS 6381 - Steel Tubing, Mechanical
0.95Cr - 0.20Mo (0.38 - 0.43C) (SAE 4140).
(DoD adopted)

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Hose. Hose furnished under this specification shall conform to the requirements of MIL-H-24135 and as specified herein.

3.2 Material.

3.2.1 Hose. Rubber compounds used shall be resistant to petroleum base oils and sea water and shall consist of synthetic rubber or a blend of synthetic rubbers and shall meet the requirements of MIL-H-24135 and as specified herein.

3.2.2 Fittings. Material for fittings shall be as shown on figures 1 through 13.

3.3 Reinforcement. Reinforcement shall consist of four plies spiral wrapped wire applied in such a manner as to meet all the requirements of this specification.

3.4 Dimensions and pressures shall be in accordance with table I.

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TABLE I. Dash numbers, performance characteristics, and dimensions.

Hose size (dash no.)	Pressure (lb/in^2)		Hose dimensions (inches)						Min bend radius (inches)	Approx wt/ft (lbs/ft)
	Working (max)	Burst (min)	Id tube		Od cover		Od reinf			
			min	max	min	max	min	max		
-16	4,000	16,000	0.985	1.040	1.515	1.609	1.360	1.422	14	1.42
-20	3,000	12,000	1.235	1.297	1.938	2.062	1.781	1.859	18	2.15
-24	2,500	10,000	1.485	1.547	2.188	2.312	2.031	2.109	22	2.47
-32	2,000	8,000	1.985	2.047	2.688	2.812	2.531	2.609	28	3.13

3.5 Fitting types and sizes. Fitting types and sizes shall be as shown on figures 1 through 13.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-H-24135 and as specified herein.

4.2 Qualification samples for testing.

4.2.1 Hose. Qualification testing of hose shall consist of hose samples as specified in table II. Samples shall be not less than 15 inches in length and shall be fitted with qualified end fittings.

TABLE II. Number of hose samples for qualification testing.

Hose size (dash no.)	Number of hose samples to be provided
-16	4
-32	5

4.2.2 End fittings. Qualification testing of end fittings only shall consist of 18 end fittings of the same type as specified in table III. End fitting samples shall be installed on nine qualified hose samples, in the sizes listed in table II.

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TABLE III. Number of end fitting samples for qualification testing.

End fitting size (hose dash no.)	Number of end fitting samples to be provided
-16	8
-32	10

4.2.3 Hose and end fittings combined. When a single contractor of both hose and fittings qualifies both hose and fittings, or a hose contractor and a fitting contractor qualifies the hose and fittings in a combined qualification test, the samples shall be provided as specified in table IV.

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TABLE IV. Number of samples for combined qualification testing.

One approved source					
Hose size (dash no.)	No. of hose samples to be provided		End fitting (hose dash no.)	No. of end fittings to be provided	
	Qualifier 2/	Approved 1/ source		Qualifier 2/	Approved 1/ source
-16	2	2	-16	4	4
-32	3	2	-32	5	5

Two or more approved sources						
Hose size (dash no.)	Number of samples to be provided by			End fitting size (dash no.)	Number of end fittings to be provided by		
	Qualifier 2/	Approved 1/ source no. 1	Approved 1/ source no. 2		Qualifier 2/	Approved 1/ source no. 1	Approved 1/ source no. 2
-16	2	1	1	-16	4	2	2
-32	3	1	1	-32	5	3	2

Hose samples shall be not less than 15 inches in length and shall be coupled on one end with the fitting being submitted for qualification and on the other end with previously qualified end fittings.

1/ Approved source - Previously qualified hose and fittings, or hose or fitting.
2/ Qualifier - Hose and fittings, or hose or fitting to be qualified.

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4.2.4 Sequence of tests on sample hose assemblies shall be as specified in table V.

TABLE V. Sequence of qualification tests on sample assemblies.

	Sample numbers								
	Size (dash no.) -16				Size (dash no.) -32				
Test sequence	1	2	3	4	5	6	7	8	9
Proof pressure (see 4.6.1)1/	X	X	X	X	X	X	X	X	X
Stability test (see 4.6.2)1/	X	X	X	X	X	X	X	X	X
Aging test (see 4.6.3)1/		X	X	X		X	X	X	X
Hydraulic fluid circulation (see 4.6.4)1/		X				X			
Impulse test (see 4.6.5)1/				X				X	X
Burst test (see 4.6.6)1/	X				X				
Cold temperature flexing (see 4.6.8)1/			X				X		

1/ See MIL-H-24135.

4.3 The test fluid shall be petroleum base oil in accordance with MIL-L-17331 or MIL-L-2104 for the aging, hydraulic fluid circulation, and impulse tests.

4.4 The peak pressure for the dynamic impulse test shall be 150 percent or maximum working pressure (see table I).

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 The packaging requirements shall be in accordance with MIL-H-24135.

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6. NOTES

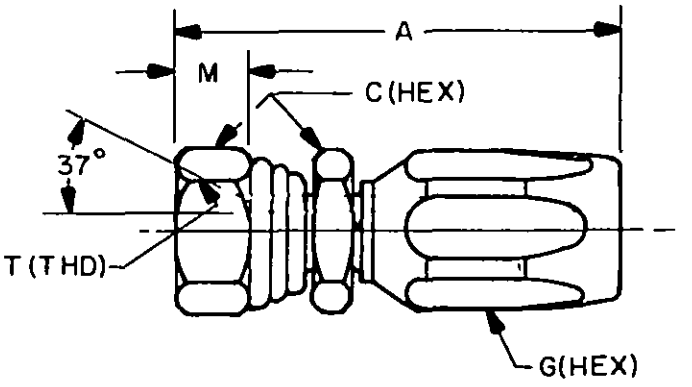
6.1 Intended use. Hose and fittings to this specification are intended for use in shipboard fluid piping systems where pressures do not exceed the rated working pressure of the hose and temperature does not exceed minus 40 to 200°F.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of end fittings for hoses (see 1.2).

Preparing activity:
Navy - SH
(Project 4720-N530-3)

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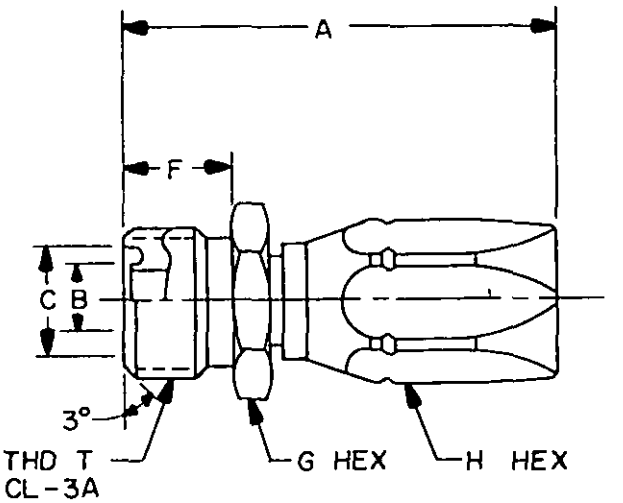
SH 1320218

Hose size (dash no.)	Tube size (nom)	T thread CL-2B	A (max)	C (see note 2)	G (see note 2)	M (ref)
-16	1	1-5/16-12 UN	5.00	1.50	2.00	0.60

NOTES:
1. Dimensions are in inches.
2. Fitting assembly materials for the swivel nut, hose nipple and socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished.
3. Socket (body) alternative material shall conform to QQ-N-288, composition B.

FIGURE 1. Type AS, 37-degree tube swivel to hose, straight.

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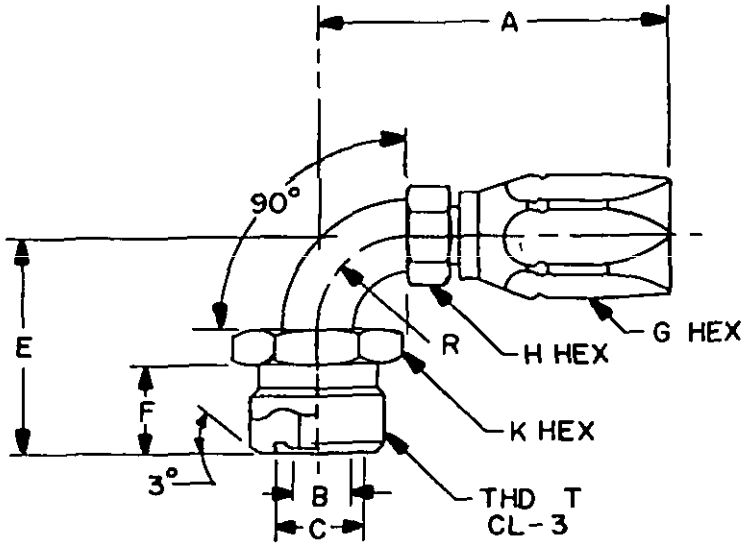
Hose size (dash no.)	Tube size (nom)	T thread CL-3	A (max)	B ± 0.005	C ± 0.005	F (ref)	G (min)	H (max)
-16	1	2-1/2-12 UN	5.400	1.093	1.375	1.12	1-3/8	2.00

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for the male half union adapter, hose nipple, and socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished.
3. Socket (body) alternative material shall conform to QQ-N-288, composition B.

FIGURE 2. Type BS, gasket seal union, male half to hose, straight.

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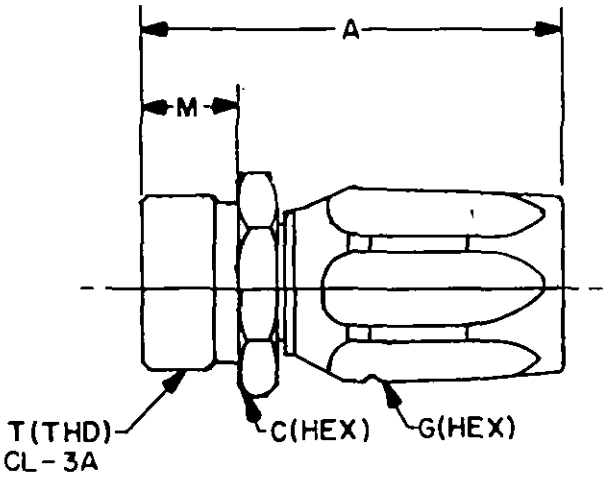
SH 1320220

Hose size (dash no.)	Tube size (nom)	T thread CL-3	A (max)	B ± 0.005	C ± 0.005	E (max)	F (ref)	G (min)	H (min)	K (min)	R
-16	1	2-1/8-12 UN	6.50	1.093	1.375	3.00	1.12	2.00	1-5/8	2-1/4	1.50

NOTES:
1. Dimensions are in inches.
2. Fitting assembly materials for the male half union adapter, hose nipple and socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished.
3. The 90-degree tube elbow shall conform to MIL-T-16420, composition 70-30, grade 2, type I, class 3300 special.
4. Socket (body) alternative material shall conform to QQ-N-288, composition B.

FIGURE 3. Type BLS, gasket seal union, male half to hose, 90-degree elbow.

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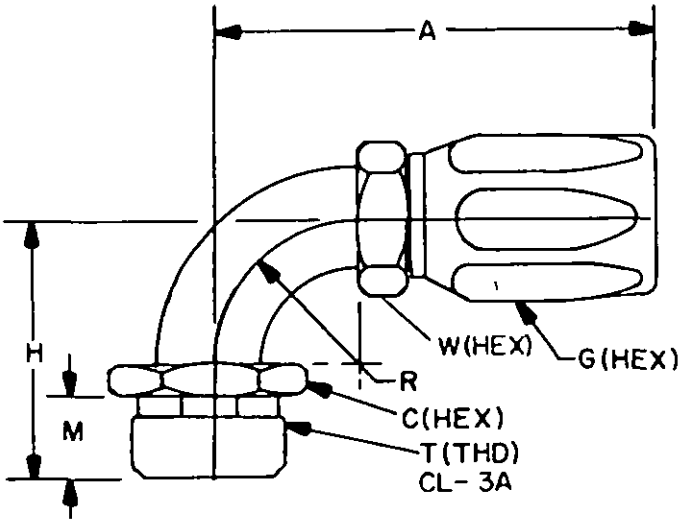
SH 1320221

Hose size (dash no.)	Tube size (nom)	T thread CL-3A	A (max)	C (min)	G (min)	M (ref)
-16	1	2-5/16-12 UN	4.75	2-3/8	2.00	0.75

NOTES:
1. Dimensions are in inches.
2. Fitting assembly materials for the male half union adapter, hose nipple and socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished.
3. Socket (body) alternative material shall conform to QQ-N-288, composition B.

FIGURE 4. Type CS, O-ring seal union, male half to hose, straight.

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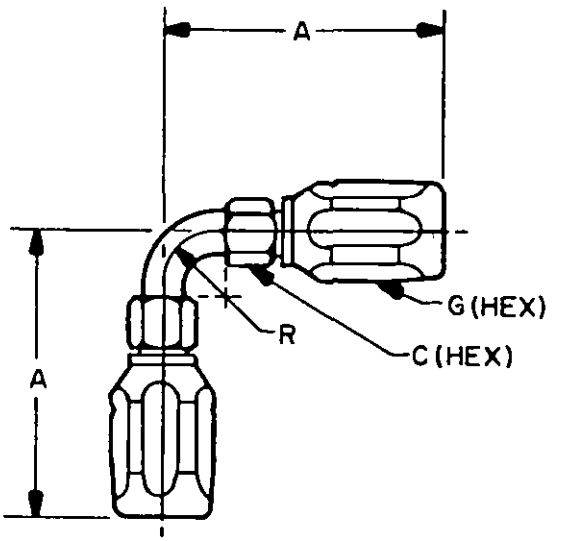
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Hose size (dash no.)	Tube size (nps)	T thread CL-3A	A (max)	C (min)	G (min)	H (max)	M (ref)	R (ref)	W (min)
-16	1	2-5/16-12 UN	6.50	2-3/8	2.00	3.50	0.75	1.50	1-3/8

NOTES:
1. Dimensions are in inches.
2. Fitting assembly materials for the male half union adapter, hose nipple and socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished.
3. The 90-degree tube elbow shall conform to MIL-T-16420, composition 70-30, grade 2, type I, class 3300 special.
4. Socket (body) alternative material shall conform to QQ-N-288, composition B.

FIGURE 5. Type CLS, O-ring seal union, male half to hose, 90-degree elbow.

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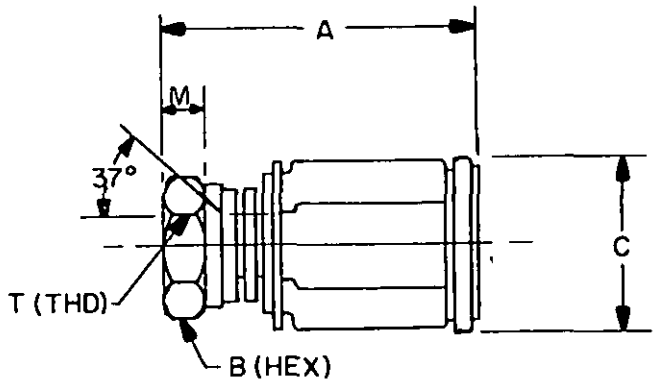
Hose size (dash no.)	A (max)	C (min)	G (min)	R (ref)
-16	6.50	1-3/8	2.00	1.50

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for the male half union adapter, hose nipple and socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished.
3. Socket (body) alternative material shall conform to QQ-N-288, composition B.
4. The 90-degree tube elbow shall conform to MIL-T-16420, composition 70-30, grade 2, type I, class 3300 special.

FIGURE 6. Type DS, 90-degree elbow, hose to hose.

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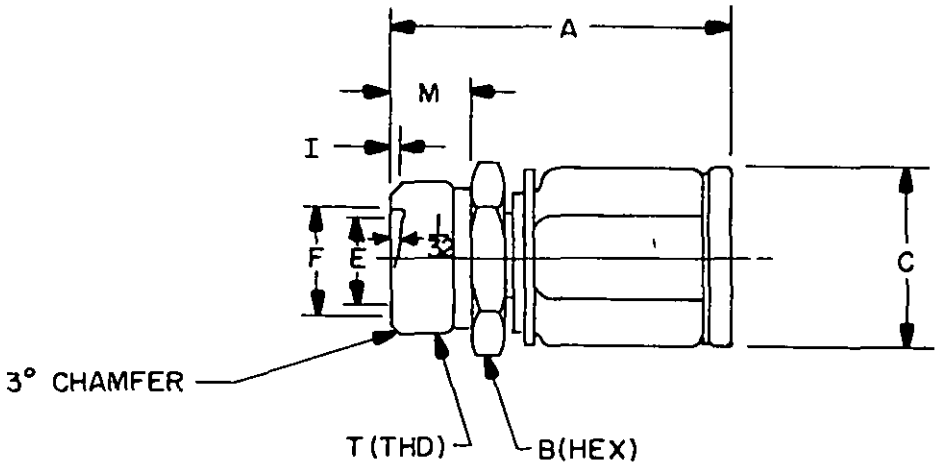
Hose size (dash no.)	Tube size (nom)	T thread CL-2B	A (max)	B (min)	C (max)	M (ref)
-20	1-1/4	1-5/8-12 UN	6.50	2	4.00	0.64
-24	1-1/2	1-7/8-12 UN	7.25	2-1/4	4.25	.74
-32	2	2-1/2-12 UN	8.25	2-7/8	5.25	.93

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for the swivel nut and hose nipple shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished.
3. Socket for fitting assemblies for -20 hose and larger may be the one-piece or multiple segment design not exceeding the envelope dimensions.
4. Socket (body) alternative material shall conform to QQ-N-288, composition B.
5. Alternative material for the fitting assembly socket shall be steel conforming to AISI 1035 (segment) or SAE 4140 (ring x-band) in accordance with AMS 6381 as applicable.

FIGURE 7. Type CLS, SAE 37-degree tube swivel to hose, straight.

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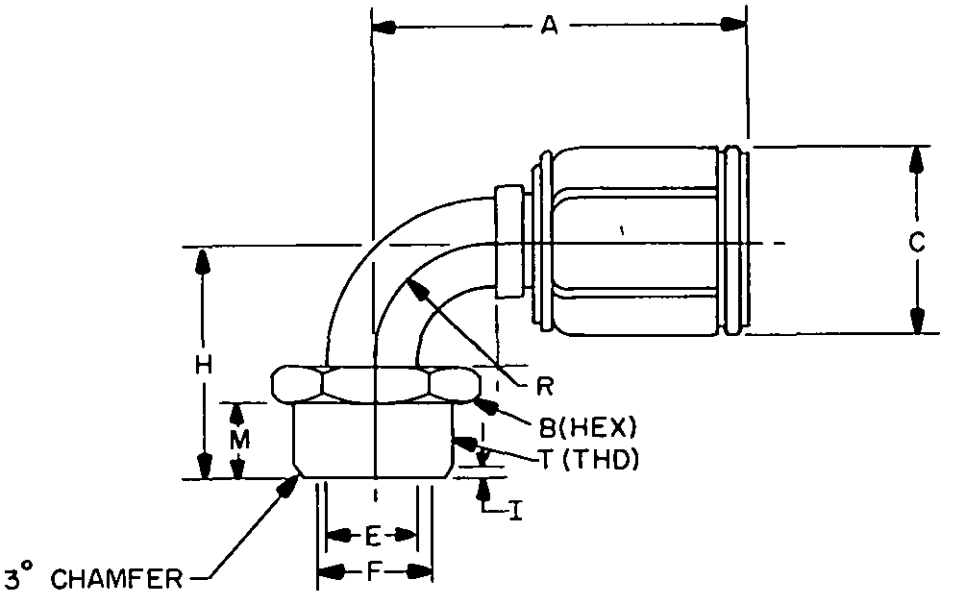
Hose size (dash no.)	Tube size (nom)	T thread CL-2B	A (max)	B (min)	C (max)	E +0.005	F +0.005	I (ref)	M (ref)
-20	1	2-1/8-12 UN	6.25	2-1/4	4.00	1.093	1.375	5/64	1.12
-20	1-1/4	2-5/8-12 UN	6.50	2-3/4	4.00	1.344	1.750	3/32	1.12
-24	1-1/4	2-5/8-12 UN	7.00	2-3/4	4.25	1.344	1.750	3/32	1.25
-24	1-1/2	2-7/8-12 UN	7.25	3	4.25	1.656	2.094	7/64	1.31
-32	2	3-5/8-12 UN	8.00	3-3/4	5.25	1.937	2.375	1/8	1.38

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for the male half union adapter and hose nipple shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished.
3. Socket for fitting assemblies for -20 hose and larger may be the one-piece or multiple segment design not exceeding the envelope dimensions.
4. Material for fitting assembly socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished or QQ-N-288, composition B.
5. Alternative material for the fitting assembly socket shall be steel conforming to AISI 1035 (segment) or SAE 4140 (ring x-band) in accordance with AMS 6381 as applicable.

FIGURE 8. Type BO, gasket seal union, male half to hose, straight.

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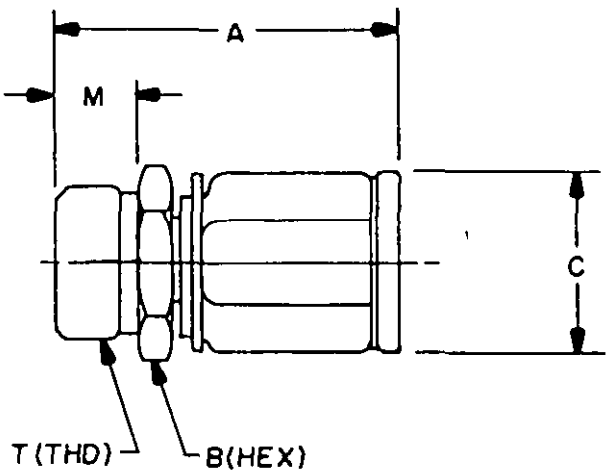
Hose size (dash no.)	Tube size (nom)	T thread CL-3	A (max)	B (min)	C (max)	E ± 0.005	F ± 0.005	H (max)	I (ref)	M (ref)	R rad (min)
-20	1-1/4	2-5/8-12 UN	7.00	2-3/4	4.00	1.344	1.750	4.75	3/32	1.25	1.75
-24	1-1/2	2-7/8-12 UN	7.75	3	4.25	1.656	2.004	5.25	7/64	1.31	2.00
-32	2	3-5/8-12 UN	7.98	3-3/4	5.25	1.937	2.375	6.00	1/8	1.38	2.50

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for the male half union adapter and hose nipple shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished.
3. Socket for fitting assemblies for -20 hose and larger may be the one-piece or multiple segment design not exceeding the envelope dimensions.
4. Material for fitting assembly socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished or QQ-N-288, composition B.
5. Alternative material for the fitting assembly socket shall be steel conforming to AISI 1035 (segment) or SAE 4140 (ring x-band) in accordance with AMS 6381 as applicable.
6. The 90-degree tube elbow shall conform to MIL-T-16420, composition 70-30, grade 2, type I, class 1650 special.

FIGURE 9. Type BLO, gasket seal union, male half to hose, 90-degree elbow.

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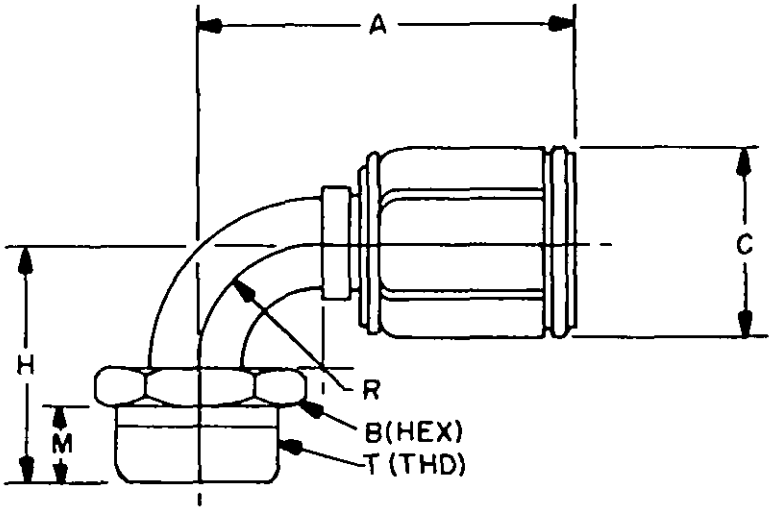
Hose size (dash no.)	Tube size (nom)	T thread CL-3A	A (max)	B (min)	C (max)	M (ref)
-20	1	2-5/16-12 UN	6.25	2.38	4.00	0.75
-20	1-1/4	2-3/4-12 UN	6.50	2.88	4.00	.75
-24	1-1/4	2-3/4-12 UN	7.00	2.88	4.25	.75
-24	1-1/2	3-1/16-12 UN	7.25	3.12	4.25	.88
-32	2	3-3/4-12 UN	8.00	3.88	5.25	.88

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for the hose nipple shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished.
3. Socket for fitting assemblies for -20 hose and larger may be the one-piece or multiple segment design not exceeding the envelope dimensions.
4. Material for fitting assembly socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished or QQ-N-288, composition B.
5. Alternative material for the fitting assembly socket shall be steel conforming to AISI 1035 (segment) or SAE 4140 (ring x-band) in accordance with AMS 6381 as applicable.

FIGURE 10. Type CO, O-ring union, male half to hose, straight.

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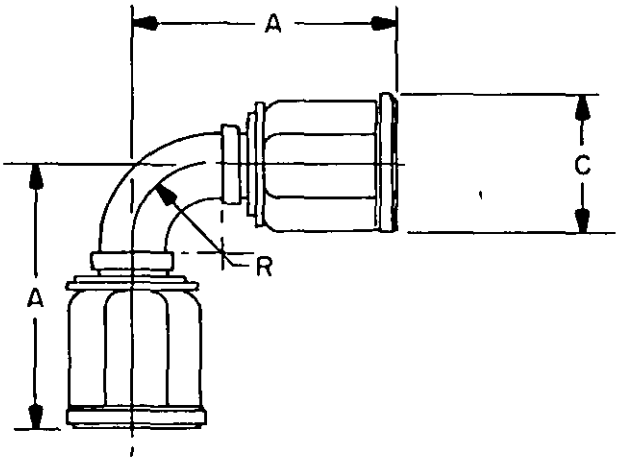
Hose size (dash no.)	Tube size (nps)	T Thread CL-3A	A (max)	B (min)	C (max)	H (max)	M (ref)	R rad (min)
-20	1-1/4	2-3/4-12 UN	7.00	2.88	4.00	4.75	0.75	1.75
-24	1-1/2	3-1/16-12 UN	7.75	3.12	4.25	5.25	.88	2.00
-32	2	3-3/4-12 UN	8.00	3.88	5.25	6.00	.88	2.50

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for the male half union adapter and hose nipple shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished.
3. Socket for fitting assemblies for -20 hose and larger may be the one-piece or multiple segment design not exceeding the envelope dimensions.
4. Material for fitting assembly socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished or QQ-N-288, composition B.
5. Alternative material for the fitting assembly socket shall be steel conforming to AISI 1035 (segment) or SAE 4140 (ring x-band) in accordance with AMS 6381 as applicable.
6. The 90-degree tube elbow shall conform to MIL-T-16420, composition 70-30, grade 2, type I, class 1650 special.

FIGURE 11. Type CLO, O-ring union, male half to hose, 90-degree elbow.

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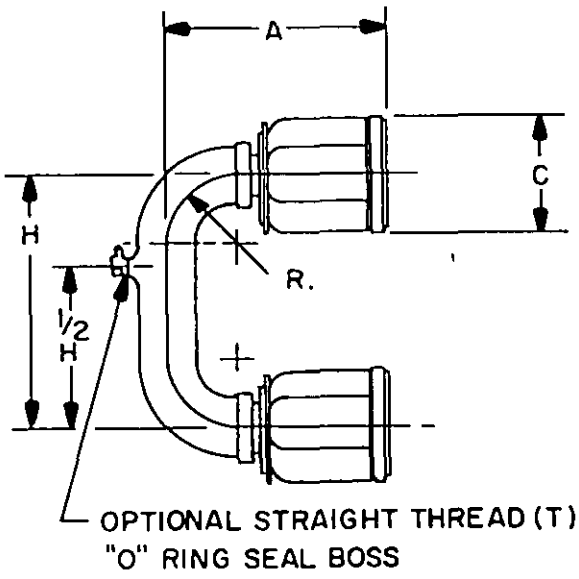
Hose size (dash no.)	A (max)	C (max)	R rad (min)
-20	7.00	4.00	1.75
-24	7.75	4.25	2.00
-32	9.00	5.25	2.50

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for hose nipple shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished.
3. Socket for fitting assemblies for -20 hose and larger may be the one-piece or multiple segment design not exceeding the envelope dimensions.
4. Material for fitting assembly socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished or QQ-N-288, composition B.
5. Alternative material for the fitting assembly socket shall be steel conforming to AISI 1035 (segment) or SAE 4140 (ring x-band) in accordance with AMS 6381 as applicable.
6. The 90-degree tube elbow shall conform to MIL-T-16420, composition 70-30, grade 2, type I, class 1650 special.

FIGURE 12. Type DO, 90-degree elbow, hose to hose.

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Hose size (dash no.)	Optional T thread UNF-2B	A (max)	C (max)	H (max)	R (min)
-24	5/16-24	7.75	4.25	6.50	2.00

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for hose nipple shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished.
3. Socket for fitting assemblies for -20 hose and larger may be the one-piece or multiple segment design not exceeding the envelope dimensions.
4. Material for fitting assembly socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished or QQ-N-288, composition B.
5. Alternative material for the fitting assembly socket shall be steel conforming to AISI 1035 (segment) or SAE 4140 (ring x-band) in accordance with AMS 6381 as applicable.
6. The 90-degree tube elbow shall conform to MIL-T-16420, composition 70-30, grade 2, type I, class 1650 special.

FIGURE 13. Type UO, 180-degree elbow, hose to hose.

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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